Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/10/2020
PI Name:	Hirsa, Amir Ph.D.		
Project Title:	Amyloid Fibril Formation in Microgravity via Ring-Sheared Drop		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	09/30/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:	8	**No. of Master' Degrees:**	6
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	20	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2022 per continuation grant 80NSSC20K1726 issued, including title change now reflected in above Title, per S. Gorti/MSFC (Ed., 1/27/21) NOTE: Extended to 9/30/2020 per PI and NSSC information (Ed., 6/4/18)		
Key Personnel Changes/Previous PI:	June 2020 report: no changes		
COI Name (Institution):	Lopez, Juan Ph.D. (Arizona State University)		
Grant/Contract No.:	80NSSC20K1726 ; NNX13AQ22G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note January 2021: Continuation grant issued with slightly different title, Amyloid Fibril Formation in Microgravity via Ring-Sheared Drop 80NSSC20K1726 (formerly titled Amyloid Fibril Formation in Microgravity: Distinguishing Interfacial and Flow Effects NNX13AQ22G). Information combined with original grant NNX13AQ22G, per NASA Marshall Space Flight Center management] Formation of amyloid fibrils is widely studied because of their role in neurodegenerative diseases, including Alzheimer's. The fibrillization process is characterized by a nucleation step followed by fibril growth, much like the steps that occur during protein crystallization. Aside from the general interest in macromolecular biophysics, recent evidence shows the relevance of amyloid fibril formation in animal models exposed to cosmic radiation, with obvious implications for interplanetary travel. A preponderance of evidence shows that agitating protein solutions can significantly accelerate amyloid fibril formation; however, the mechanisms by which the presence of bubbles and flow independently affect fibrillization are not known. The situation is further complicated because we do not yet have mechanistic models that can quantify the influence of natural convection due to gravity on the fibrillization process. We know that fluid interfaces (e.g., air-water interface) induce amyloid fibrils when they are sheared, but the influence of container walls is not fully understood. Here, we aim to distinguish the influence of flow and fluid interfaces on amyloid formation by removing other effects, including solid walls and buoyancy effects. Formation of protein amyloid fibrils provides a model system for studying biomolecular transport phenomena, where the influence of flow can be altogether negated by the elimination of gravity. Furthermore, the experiments can be performed aboard the International Space Station (ISS) with essentially no solid container, since surface tension can provide liquid containment. Thus, the free surface can be readily observed (e.g., with high magnification objectives), manipulated (e.g., can be sheared or left otherwise undisturbed), and modified (e.g., by spreading a monomolecular oil film). First, we will perform experiments at 1g on a small droplet to ascertain the effect of forced convection on a liquid surface. Next, we will generalize our very successful computational model in annular systems to the spherical geometry of liquids at microgravity. We will design the space-based experiments based on the results from the ground experiments and numerical predictions. Finally, we will perform experiments aboard ISS on large-scale drops in an experiment referred to as the Ring-Sheared Drop (RSD).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	When proteins are perturbed from their native state, they lose their native confirmation (folding) and then can bond with other proteins, forming an aggregate. This is the reason why the transparent fluid in an egg becomes a white solid when cooked. Amyloid fibrils are a unique type of protein aggregate that occur naturally in the body. We study them because amyloid plaques in the brain are associated with many neurodegenerative disorders, including Alzheimer's. Fibrils are very intriguing because they are highly ordered and self-assemble. When heated, insulin molecules, for example, denature and bond with other denatured molecules (or assists in denaturing the other molecules). These form a nucleation site, and from there the fibril elongates as more molecule bond into what is called a "beta sheet" structure. The primary mechanism for activation of protein in the body is not yet understood, but it is likely to be a combination of flow friction, aka "shear stress," and interfacial effects that native proteins are exposed to. Aside from their central role in neurodegenerative diseases such as Parkinson's and Alzheimer's, formation of amyloid structures in proteins plays a role in other diseases including diabetes.
Task Progress:	Scientific and Technological Progress A) A set of 6 protein flight samples was prepared, including degassing and deep-freezing, and was subsequently shipped to NASA. Two types of protein samples were prepared: native and pre-sheared. Native protein solution was pre-sheared in deep-channel surface viscometers. Two of the samples, one native and one pre-sheared, included controls to evaluate the effect of the thawing and subsequent refrigeration but without drop deployment in the RSD. Additional samples were made and stored at the same temperature in our lab for subsequent comparisons to post-flight samples. For quality control, a comprehensive battery of tests was performed on the protein solutions including soluble protein concentration, Thioflavin-T fluorescence, circular-dichroism, and fluorescence microscopy. B) A system for remote operation of the ISS experiment was set up at RPI (Rensselaer Polytechnic Institute). A team of four individuals were also trained for conducting space ops. C) Two calibration runs were made by NASA-Marshall Space Flight Center (MSFC) and Teledyne–Brown Engineering ("TBE," a NASA contractor), in which RPI team assisted. The first utilized de-ionized water, which successfully deployed and pinned. The drop was sheared at 30 rpm, the maximum speed of the instrument. D) Following the space ops, a set of protein samples returned from space were analyzed at RPI and compared to Earth samples. Collectively, the analysis showed that native protein will remain in the native state throughout the degassing, freezing, thawing, refrigerated storage, space travel and transportation phases of the experiment. This supports the validity of the experimental design. It is also observed that the presheared protein is in the expected state of protein structure and fibrillization progression. Pre-sheared space samples were both in the final stages of either fibrillization or aggregation. One of the space samples was fully fibrillized, and the other was partially fibrillized with a large amount of random aggregates. It is also important to note that the analysis shows that even without shearing in the RSD, protein solution treated with pre-shearing protocol will fibrillize or aggregate. Given this information, it may be much more useful in the future to use native protein only, to ensure fibrillization results are due to shear in the RSD. Another option could also be to shorten the time protein was pre-sheared for. Overall the results are informative and useful for the continuation and improvement of future RSD experiments. E) Extensive experimental studies were conducted to simulate microgravity in the RSD at the scale of the ISS module, to examine aspects of drop deformation. The results of the study, which utilized silicone oil in a density-matched saline solution, was published recently (McMackin, P.M., Griffin, S.R., Riley, F.P, Gulati, S., Debono, N.E., Raghunandan, A., Lopez, J.M., and Hirsa, A.H. "Simulated microgravity in the ring-sheared drop," Nature Microgravity 6, 2, 2020). The study included simulations of non-deformed drops using COMSOL software. F) A new study of drop deformation in the RSD consisting of a combination of asymptotic analysis, numerical simulations, and Earth-based bound experiments is underway to examine the mechanics of drop deformation due to the induced meridional flow. In particular, we are determining how drop deformation depends on the Reynolds number (i.e., how fast the ring is rotating compared to the viscous time scale) and the Capillary number (i.e., ratio of inertial to surface tension forces). The experiments involve a high-viscosity silicone drop surrounded by a density matched salt-water solution. The large ratio of the dynamic viscosities of the two liquids (approximately 100) simulates the RSD experiment aboard the ISS, where the flow in the surrounding fluid (air in the case of the ISS, with a viscosity ratio of approximately 60) is decoupled from the flow inside the

drop (protein in water). Good agreement has been found between the theory and experiments, and the results are under preparation for journal submission.

G) The application of the RSD as a viable space-based bioreactor was studied and verified using its Earth-based analog, the knife-edge surface viscometer. The results showed that E. coli could be grown and green fluorescence protein could be expressed in E. coli as efficiently in the knife-edge surface viscometer as in standard laboratory orbital shakers. The results were published recently (Adam, J.A., Gulati, S., Hirsa, A.H., and Bonocora, R.P. “Growth of microorganisms in an interfacially driven space bioreactor analog,” Nature Microgravity 6, 11, 2020).

H) Throughout much of the past few months, the Principal Investigator and his team have also supported the engineering effort by MSFC and TBE. Extensive experimentation was undertaken at RPI to test modifications to the D-tube to ensure proper pinning and deployment of protein solutions in the RSD module. These efforts are ongoing and we are expanding our research effort in this area.

Bibliography Type:	Description: (Last Updated: 04/15/2026)
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